PROGRAM 8
NODE.JS APPLICATION USING CONFIGMAP AND POD IN KUBERNETES

STEP 1: Create server.js

```
const http = require('http');
const port = 3000;

const server = http.createServer((req, res) => {
  res.writeHead(200, { "Content-Type": "text/plain" });
  res.end("Hello from Node.js running inside Kubernetes!\n");
});

server.listen(port, () => {
  console.log(`Server is running on port ${port}`);
});
```

STEP 2: Create ConfigMap YAML

Filename: nodejs-configmap.yaml

```
apiVersion: v1
kind: ConfigMap
metadata:
  name: nodejs-app-config
data:
  server.js: |
    const http = require('http');
    const port = 3000;

    const server = http.createServer((req, res) => {
      res.writeHead(200, { "Content-Type": "text/plain" });
      res.end("Hello from Node.js running inside Kubernetes!\n");
    });

    server.listen(port, () => {
      console.log(`Server is running on port ${port}`);
    });
```

STEP 3: Create Pod YAML

Filename: nodejs-pod.yaml

```
apiVersion: v1
kind: Pod
```

```yaml
metadata:
  name: program-8
  labels:
    app: nodejs
spec:
  containers:
    - name: nodejs
      image: node:18
      command: ["node", "/usr/src/app/server.js"]
      ports:
        - containerPort: 3000
      volumeMounts:
        - name: app-code
          mountPath: /usr/src/app
  volumes:
    - name: app-code
      configMap:
        name: nodejs-app-config
```

STEP 4: Apply ConfigMap and Pod

```
kubectl apply -f nodejs-configmap.yaml
kubectl apply -f nodejs-pod.yaml
```

STEP 5: Check Pod Status

```
kubectl get pods
kubectl describe pod program-8
kubectl logs program-8
```

STEP 6: Access Node.js Application

Port-forward:

```
kubectl port-forward pod/program-8 3000:3000
```

Open in browser:

http://localhost:3000

STEP 7: Cleanup

```
kubectl delete pod program-8
kubectl delete configmap nodejs-app-config
```

OUTPUT:

Node.js application successfully executed inside Kubernetes using ConfigMap and Pod.

```
1RV24MC025_CHANDAN_K@chandan-Vivobook-ASUSLaptop-M6500QC-M6500QC:~$ mkdir prog8
1RV24MC025_CHANDAN_K@chandan-Vivobook-ASUSLaptop-M6500QC-M6500QC:~$ cd prog8
1RV24MC025_CHANDAN_K@chandan-Vivobook-ASUSLaptop-M6500QC-M6500QC:~/prog8$ nano server.js
1RV24MC025_CHANDAN_K@chandan-Vivobook-ASUSLaptop-M6500QC-M6500QC:~/prog8$ nano nodejs-configmap.yaml
1RV24MC025_CHANDAN_K@chandan-Vivobook-ASUSLaptop-M6500QC-M6500QC:~/prog8$ cat nodejs-configmap.yaml
apiVersion: v1
kind: ConfigMap
metadata:
  name: nodejs-app-config
data:
  server.js: |
    const http = require('http');
    const port = 3000;

    const server = http.createServer((req, res) => {
      res.writeHead(200, { "Content-Type": "text/plain" });
      res.end("Hello from Node.js running inside Kubernetes!\n");
    });

    server.listen(port, () => {
      console.log(`Server is running on port ${port}`);
    });
1RV24MC025_CHANDAN_K@chandan-Vivobook-ASUSLaptop-M6500QC-M6500QC:~/prog8$ nano nodejs-pod.yaml
1RV24MC025_CHANDAN_K@chandan-Vivobook-ASUSLaptop-M6500QC-M6500QC:~/prog8$ kubectl apply -f nodejs-configmap.yaml
configmap/nodejs-app-config created
1RV24MC025_CHANDAN_K@chandan-Vivobook-ASUSLaptop-M6500QC-M6500QC:~/prog8$ kubectl apply -f nodejs-pod.yaml
pod/program-8 created
1RV24MC025_CHANDAN_K@chandan-Vivobook-ASUSLaptop-M6500QC-M6500QC:~/prog8$ kubectl get pods
NAME                                READY   STATUS              RESTARTS        AGE
kubia                               1/1     Running             1 (9m6s ago)    3d23h
nginx-deployment-6f9664446b-bml6q   1/1     Running             0               8m11s
nginx-deployment-6f9664446b-rcf5d   1/1     Running             0               8m11s
program-8                           0/1     ContainerCreating   0               26s
test-6bb654b8f8-p82wd               1/1     Running             1 (3d14h ago)   3d23h
1RV24MC025_CHANDAN_K@chandan-Vivobook-ASUSLaptop-M6500QC-M6500QC:~/prog8$ kubectl port-forward pod/program-8 3000:3000
error: unable to forward port because pod is not running. Current status=Pending
1RV24MC025_CHANDAN_K@chandan-Vivobook-ASUSLaptop-M6500QC-M6500QC:~/prog8$ kubectl port-forward pod/program-8 6000:6000
Forwarding from 127.0.0.1:6000 -> 6000
Forwarding from [::1]:6000 -> 6000
1RV24MC025_CHANDAN_K@chandan-Vivobook-ASUSLaptop-M6500QC-M6500QC:~/prog8$ kubectl describe pod program-8
Name:         program-8
Namespace:    default
Priority:     0

1RV24MC025_CHANDAN_K@chandan-Vivobook-ASUSLaptop-M6500QC-M6500QC:~/prog8$ kubectl logs program-8
Server is running on port 3000
1RV24MC025_CHANDAN_K@chandan-Vivobook-ASUSLaptop-M6500QC-M6500QC:~/prog8$ kubectl port-forward pod/program-8 3000:3000
Forwarding from 127.0.0.1:3000 -> 3000
Forwarding from [::1]:3000 -> 3000
Handling connection for 3000
```